

An improved spectral averaging bound for the random Toeplitz limit law

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Abstract

Let $T_n = (a_{|i-j|})_{1 \leq i, j \leq n}$ be the random symmetric Toeplitz matrix with independent centered entries of variance one, and let γ_T be the limiting spectral law of $n^{-1/2}T_n$ introduced by Bryc, Dembo and Jiang. Sen and Virág proved that γ_T is absolutely continuous with bounded density. We give a short spectral-averaging proof of the explicit bound

$$\|\rho_T\|_\infty \leq \pi^{-1/2}.$$

More precisely, the same bound holds uniformly for the expected empirical spectral measures of the Gaussian $\sqrt{2}$ -modified finite Toeplitz ensemble. The proof keeps the circulant embedding of Sen–Virág but replaces the Wegner-type averaging estimate by an exact rank-two Krein/Poisson computation: each two-dimensional Fourier channel contributes exactly its Poisson mass.

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1 Introduction

Let $(a_j)_{j \geq 0}$ be independent real random variables with $\mathbb{E}a_0 = 0$ and $\mathbb{E}a_0^2 = 1$, and set

$$T_n = (a_{|i-j|})_{1 \leq i, j \leq n}.$$

Bryc, Dembo and Jiang proved that the empirical spectral distribution of $n^{-1/2}T_n$ converges weakly, almost surely, to a nonrandom symmetric probability measure γ_T that is universal in the entry distribution [1]. They also proved that γ_T has unbounded support and expressed its even moments as sums of Toeplitz volumes indexed by pair partitions.

The regularity of γ_T has been a persistent question since the original paper. Sen and Virág proved that γ_T is absolutely continuous with bounded density by combining a circulant embedding with a spectral averaging estimate from random Schrödinger operator theory [3]. The purpose of this note is to give a short refinement of the averaging step. The new input is an exact Poisson integral for the rank-two perturbations associated with the paired Fourier modes of the embedding.

We state the result in a finite- n form. Throughout the proof the entries are standard Gaussian. Define the $\sqrt{2}$ -modified Toeplitz matrix T_n° by replacing the diagonal entry a_0 of T_n by $\sqrt{2}a_0$:

$$(T_n^\circ)_{ij} = a_{|i-j|} \quad (i \neq j), \quad (T_n^\circ)_{ii} = \sqrt{2}a_0.$$

Let μ_A denote the empirical spectral measure of a Hermitian matrix A .

Theorem 1.1 (Improved density bound). *For every $n \geq 1$, the expected empirical spectral measure*

$$\nu_n^\circ := \mathbb{E} \mu_{n^{-1/2} T_n^\circ}$$

has a density ρ_n° satisfying

$$\|\rho_n^\circ\|_\infty \leq \pi^{-1/2}.$$

Consequently the Toeplitz limiting law γ_T is absolutely continuous and has a density ρ_T satisfying

$$\|\rho_T\|_\infty \leq \pi^{-1/2}.$$

The modification of the diagonal does not change the limiting law: T_n° differs from T_n by the scalar matrix $(\sqrt{2} - 1)a_0 I_n$, so after division by $\sqrt{n}$ the spectrum is shifted by a random scalar tending to zero almost surely. Since the Gaussian Toeplitz ensemble has the same limit law γ_T by universality [1], the finite- n estimate implies the assertion for γ_T .

2 Circulant embedding and the Stieltjes identity

For a probability measure ν on $\mathbb{R}$ define its Stieltjes transform by

$$s(z; \nu) = \int_{\mathbb{R}} \frac{1}{x - z} \nu(dx), \quad \Im z > 0.$$

Thus $\Im s(z; \nu) > 0$ for $\Im z > 0$.

We recall the Gaussian circulant embedding used by Sen and Virág [3]. Let $C_{2n} = (b_{(j-i) \bmod 2n})_{0 \leq i, j \leq 2n-1}$ be the real symmetric circulant with

$$b_0 = \sqrt{2}a_0, \quad b_n = \sqrt{2}a_n, \quad b_j = b_{2n-j} = a_j \quad (0 < j < n).$$

Then T_n° is the upper-left $n \times n$ corner of C_{2n} . Let $U = U_{2n}$ be the unitary discrete Fourier transform,

$$U(j, \ell) = (2n)^{-1/2} e^{2\pi i j \ell / (2n)}.$$

There is a real diagonal matrix $D = \text{diag}(d_0, \dots, d_{2n-1})$ such that

$$(2n)^{-1/2} C_{2n} = U D U^*,$$

where $d_j = d_{2n-j}$, the variables $d_0, \dots, d_n$ are independent centered Gaussians, $\text{Var}(d_j) = 1$ for $0 < j < n$, and $\text{Var}(d_0) = \text{Var}(d_n) = 2$.

Let $Q = \text{diag}(I_n, 0_n)$ and set

$$P := U^* Q U.$$

Then P is an orthogonal projection of rank n . Its matrix entries satisfy

$$P(j, j) = \frac{1}{2} \quad (0 \leq j < 2n), \quad P(j, 2n - j) = 0 \quad (j \notin \{0, n\}).$$

Indeed,

$$P(j, \ell) = \frac{1}{2n} \sum_{m=0}^{n-1} e^{2\pi i m(\ell-j)/(2n)}.$$

The diagonal formula is immediate, and the second identity follows from the sum of the nontrivial n th roots of unity.

Lemma 2.1 (Stieltjes-transform identity). *Let $z \in \mathbb{C}$ with $\Im z > 0$ and put $w = z/\sqrt{2}$. Then*

$$s(z; \nu_n^\circ) = \frac{1}{\sqrt{2}n} \sum_{j=0}^{2n-1} \mathbb{E} \left\langle P e_j, (PDP - w)^{-1} P e_j \right\rangle,$$

where PDP is viewed as a Hermitian operator on $\text{Ran } P$.

Proof. The matrix $n^{-1/2} T_n^\circ$ is unitarily equivalent to the compression of $\sqrt{2}D$ to the subspace $\text{Ran } P$. Hence

$$s(z; \nu_n^\circ) = \frac{1}{n} \mathbb{E} \text{Tr}_{\text{Ran } P} (\sqrt{2} PDP - z)^{-1} = \frac{1}{\sqrt{2}n} \mathbb{E} \text{Tr}_{\text{Ran } P} (PDP - w)^{-1}.$$

Since

$$\sum_{j=0}^{2n-1} (P e_j)(P e_j)^* = P$$

as an operator on $\text{Ran } P$, the trace on $\text{Ran } P$ is

$$\text{Tr}_{\text{Ran } P} (PDP - w)^{-1} = \sum_{j=0}^{2n-1} \left\langle P e_j, (PDP - w)^{-1} P e_j \right\rangle.$$

Taking expectations gives the formula. □

3 Exact finite-rank spectral averaging

The following Schur-complement identity is the finite-dimensional Krein formula we need.

Lemma 3.1 (Krein compression identity). *Let H_0 be Hermitian on a finite-dimensional Hilbert space, let Π be an orthogonal projection of rank r , and let X be Hermitian on $\text{Ran } \Pi$. Put*

$$H = H_0 + \Pi X \Pi.$$

Fix w with $\Im w > 0$ and set

$$C_0 = [\Pi(H_0 - w)^{-1} \Pi]_{\text{Ran } \Pi}, \quad W = C_0^{-1}.$$

Then C_0 is invertible, $\Im W < 0$ as a quadratic form, all eigenvalues of W lie in the open lower half-plane, and

$$[\Pi(H - w)^{-1} \Pi]_{\text{Ran } \Pi} = (W + X)^{-1}.$$

Proof. Decompose the space as $\text{Ran } \Pi \oplus \ker \Pi$ and write

$$H_0 - w = \begin{pmatrix} A & B \\ B^* & E \end{pmatrix}.$$

Since $\Im(H_0 - w) = -(\Im w)I$, the block E is invertible. The Schur complement formula gives

$$[\Pi(H - w)^{-1}\Pi]_{\text{Ran } \Pi} = (A + X - BE^{-1}B^*)^{-1}.$$

For $X = 0$ this equals C_0 , so $W = A - BE^{-1}B^*$ and the displayed formula follows.

If $0 \neq \varphi \in \text{Ran } \Pi$, then

$$\Im \langle \varphi, C_0 \varphi \rangle = \Im \langle \varphi, (H_0 - w)^{-1} \varphi \rangle = (\Im w) \left\| (H_0 - w)^{-1} \varphi \right\|^2 > 0.$$

Thus $\Im C_0 > 0$. With $\psi = C_0 \varphi$ this gives

$$\Im \langle \psi, W \psi \rangle = -\Im \langle \varphi, C_0 \varphi \rangle < 0,$$

so $\Im W < 0$. The numerical range of W is contained in the open lower half-plane; therefore every eigenvalue of W lies there. $\square$

Lemma 3.2 (Exact Poisson averaging). *Let W be an $r \times r$ complex matrix all of whose eigenvalues lie in the open lower half-plane. Then, for every $s \in \mathbb{R}$,*

$$\text{Tr } \Im(W + s)^{-1} > 0,$$

and

$$\int_{\mathbb{R}} \text{Tr } \Im(W + s)^{-1} ds = r\pi.$$

Consequently, for every probability density g on $\mathbb{R}$ and every $c > 0$,

$$\int_{\mathbb{R}} g(\lambda) \text{Tr } \Im(W + c\lambda)^{-1} d\lambda \leq \frac{r\pi \|g\|_{\infty}}{c}.$$

Proof. Let $\mu_1, \dots, \mu_r$ be the eigenvalues of W , counted with algebraic multiplicity. By Schur triangularization,

$$\text{Tr}(W + s)^{-1} = \sum_{k=1}^r \frac{1}{s + \mu_k}.$$

Since $\Im \mu_k < 0$,

$$\Im \frac{1}{s + \mu_k} = \frac{-\Im \mu_k}{(s + \Re \mu_k)^2 + (\Im \mu_k)^2} > 0.$$

Each summand is a Poisson kernel of total mass π , so the trace integrates to $r\pi$. The last assertion follows from positivity, the bound $g \leq \|g\|_{\infty}$, and the change of variables $s = c\lambda$. $\square$

We also record the elementary Stieltjes-transform criterion that converts the uniform resolvent bound into a density bound.

Lemma 3.3 (From Stieltjes bounds to density bounds). *Let ν be a probability measure on $\mathbb{R}$. If*

$$\sup_{\Im z > 0} \Im s(z; \nu) \leq K,$$

then ν is absolutely continuous and its density is bounded by K/π .

Proof. For $\eta > 0$,

$$\frac{1}{\pi} \Im s(E + i\eta; \nu) = \int_{\mathbb{R}} \frac{1}{\pi} \frac{\eta}{(x - E)^2 + \eta^2} \nu(dx)$$

is the convolution of ν with the Cauchy kernel. If $f \in C_c(\mathbb{R})$ is nonnegative, then this convolution converges weakly to ν , so

$$\int f d\nu = \lim_{\eta \downarrow 0} \int_{\mathbb{R}} f(E) \frac{1}{\pi} \Im s(E + i\eta; \nu) dE \leq \frac{K}{\pi} \int_{\mathbb{R}} f(E) dE.$$

By regularity, $\nu \leq (K/\pi)$ times Lebesgue measure, which proves the claim. $\square$

4 Proof of the main theorem

Proof of Theorem 1.1. Fix n and z with $\Im z > 0$. Put $w = z/\sqrt{2}$ and

$$R = (PDP - w)^{-1},$$

where PDP is restricted to $\text{Ran } P$.

We first average over a paired Fourier coordinate. Fix $0 < j < n$ and condition on all variables d_ℓ except d_j ; recall that $d_j = d_{2n-j}$. As a function of $\lambda = d_j$, the operator $H(\lambda) = PDP$ has the form

$$H(\lambda) = H_0 + \lambda(uu^* + vv^*), \quad u = Pe_j, \quad v = Pe_{2n-j},$$

where H_0 is independent of λ . The projection identities above give

$$\|u\|^2 = \|v\|^2 = \frac{1}{2}, \quad \langle u, v \rangle = P(j, 2n-j) = 0.$$

Thus $\hat{u} = \sqrt{2}u$ and $\hat{v} = \sqrt{2}v$ are orthonormal. If Π is the orthogonal projection onto their span, then

$$uu^* + vv^* = \frac{1}{2}\Pi, \quad H(\lambda) = H_0 + \frac{\lambda}{2}\Pi.$$

By Lemma 3.1, conditionally on the other variables,

$$[\Pi R \Pi]_{\text{Ran } \Pi} = \left(W_j + \frac{\lambda}{2} I_2 \right)^{-1}$$

for a 2×2 matrix W_j whose eigenvalues lie in the lower half-plane and which is independent of λ . Therefore

$$\langle u, Ru \rangle + \langle v, Rv \rangle = \frac{1}{2} \text{Tr} \left(W_j + \frac{\lambda}{2} I_2 \right)^{-1}.$$

The variable $\lambda = d_j$ is $N(0, 1)$, so its density has $\|g\|_\infty = (2\pi)^{-1/2}$. Applying Lemma 3.2 with $r = 2$ and $c = 1/2$ gives the conditional bound

$$\mathbb{E}_{d_j} \Im(\langle u, Ru \rangle + \langle v, Rv \rangle) \leq \frac{1}{2} \cdot \frac{2\pi(2\pi)^{-1/2}}{1/2} = \sqrt{2\pi}.$$

Averaging over the remaining coordinates preserves this estimate.

Next consider the two edge coordinates $j \in \{0, n\}$. Here d_j appears in a single diagonal slot, has law $N(0, 2)$, and its density is bounded by $(4\pi)^{-1/2}$. With $u = Pe_j$ and $\hat{u} = \sqrt{2}u$, the same argument at rank one gives

$$\mathbb{E}_{d_j} \Im \langle u, Ru \rangle \leq \frac{1}{2} \cdot \frac{\pi(4\pi)^{-1/2}}{1/2} = \frac{\sqrt{\pi}}{2}.$$

Now group the $2n$ terms in [Lemma 2.1](#): there are $n - 1$ paired channels and two edge channels. Hence

$$\begin{aligned} \Im s(z; \nu_n^\circ) &\leq \frac{1}{\sqrt{2}n} \left((n-1)\sqrt{2\pi} + 2 \cdot \frac{\sqrt{\pi}}{2} \right) \\ &< \sqrt{\pi}. \end{aligned}$$

By [Lemma 3.3](#), ν_n° has density bounded by $\sqrt{\pi}/\pi = \pi^{-1/2}$.

It remains to pass to the limit. The measures ν_n° converge weakly to γ_T : this follows from the Bryc–Dembo–Jiang convergence theorem for the Gaussian Toeplitz ensemble and from the fact that replacing $a_0 I_n$ by $\sqrt{2}a_0 I_n$ shifts every eigenvalue by $(\sqrt{2} - 1)a_0/\sqrt{n} \rightarrow 0$ almost surely. Therefore, for every open interval I ,

$$\gamma_T(I) \leq \liminf_{n \rightarrow \infty} \nu_n^\circ(I) \leq \pi^{-1/2}|I|.$$

By regularity, γ_T is dominated by $\pi^{-1/2}$ times Lebesgue measure. This proves absolute continuity and the stated density bound. $\square$

Remark 4.1 (Where the constant comes from). *The proof uses no smoothing in the spectral parameter. Each non-edge Fourier mode creates a rank-two perturbation with two Poisson kernels, each of total mass π . The only inequality after the exact integral is the bound of the one-dimensional Gaussian density by its maximum. Improving the constant by the same method would require information on how the random matrix W_j positions its Poisson mass relative to the Gaussian weight.*

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used generative AI tools to assist with drafting, editing, literature-check organization, and proof-audit notes. The author reviewed and edited the output, checked the mathematical claims and bibliographic information to the extent reported in the manuscript, and takes full responsibility for the content of the work.

Declaration of competing interest

The author declares no competing interests.

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